

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

Name of the Examination: **B. Tech. 4th Semester, Electrical Engg.**
 Subject: **Power Electronics-I**
 Time allowed: **3 Hrs.**

Course No: **EEPC210**
 Max. Marks: **50**

Roll No. _____

No. of questions to be attempted: **5**

- (a) Draw and explain turn-on, turn-off, gate-cathode characteristics of SCR. (4)
- (b) Draw and discuss switching characteristics of power MOSFET (3)
- Q1(c) An SCR fed from 200 V dc source is feeding a series connected pure L-C circuit. What will be the voltage across capacitor when the SCR gets commutated. (3)
- Q2(a) Discuss difficulties in parallel connection of S.C.R's. Discuss circuit to equalize turn-on and turn-off time of series connected SCRs. Hence derive expression for value of these circuit elements. (6)
- Q2(b) The average load voltage of a single phase half wave converter supplying power to R-L load ($R=10\ \Omega$ and $L=0.05\text{ H}$) with free wheeling diode is 70 V. Find firing angle in degree if source voltage is $325 \sin(314t)$ (4)
- Q3(a) A resistance heating load is controlled from a single phase supply using an AC controller. Determine the firing delay and the input power factor when the load power is at (i) 75%, (ii) 50%, (iii) 25% of its maximum. (6)
- Q3(b) The output voltage waveform of a single phase full wave rectifier can be clipped at $0.5 V_m$. Find the ratio of the rms value of the unclipped output waveform to rms value of the clipped waveform. (4)
- Q4(a) Draw circuit and waveform of three phase to single phase cycloconverter using half wave circulating type configuration for R-L load for output frequency $1/3^{\text{rd}}$ of input frequency. Draw the following waveforms for one cycle of output voltage with firing angle of P-converter as 90 degree:
- Input voltage, Output voltage of positive and negative converters.
 - Load voltage waveform.
 - Voltage waveform across CCLR.
 - Current waveform through CCLR. (7)
- Q4(b) A single phase half wave uncontrolled rectifier is fed from $300 \sin(\omega t)$ ac source and is supplying power to series connected R-E load of value $R=10\ \Omega$ and $E=100\text{ V}$. Find peak inverse voltage coming across diode. (3)
- Q5(a) Discuss the effect of source inductance on the performance of single phase full wave fully controlled converter feeding R-L load. Derive the expression for output voltage of such a converter. (6)
- Q5(b) Find the average output voltage if the SCRs of three phase full bridge fully controlled converter fed from $566 \sin(314t)$ 3-phase ac supply are fired at 45 degree. If the load current is 10 A and source reactance per phase is $0.8\ \Omega$, find the average output voltage and overlap angle in degrees. (4)
- Q6(a) Draw circuit and discuss operation of three phase half wave circulating type dual converter operating at $\alpha_{p,n}=90$ degree for R-L load. Draw the waveforms for following:
- Input voltage, Output voltage of positive and negative converters.
 - Load voltage waveform.
 - Voltage waveform across CCLR.
 - Current waveform through CCLR. (7.5)
- Q6(b) Discuss working of SCR using two transistor analogy. (2.5)
- Q7(a) Assuming that the load current is continuous for all of the following uncontrolled rectifiers, find the ripple frequency and ripple factor in output voltage:
- Single phase half wave rectifier
 - Single phase full wave rectifier
 - Three phase half wave rectifier
 - Three phase full wave rectifier (8)
- Q7(b) What are different surge ratings of SCR, discuss. (2)
- Q8(a) A 3-phase full wave fully controlled converter is feeding power to dc load at a constant current of 150 A. Draw three half cycles of input voltage; corresponding output voltage, input ac source current, current through any one SCR. Hence derive expression of rms current through any one SCR and find its value for this case. (6)
- Q8(b) Classify power devices under the category of current and voltage controlled devices, hence draw nature of their transfer characteristics. (4)